

5.2.1 Line Integrals Worksheet

1. Let C be the curve in $\mathbb{R}^3$ parameterized by $\mathbf{r}(t) = \langle \sin(t), 2t, \cos(t) \rangle$ for $0 \leq t \leq \frac{\pi}{2}$.

- a) Evaluate the integral $\int_C x^2 z \, ds$.

$$\begin{aligned} \int_C x^2 z \, ds &= \int_0^{\pi/2} \sin^2(t) \cos(t) \sqrt{\cos^2(t) + 4 + \sin^2(t)} \, dt = \int_0^{\pi/2} \sin^2(t) \cos(t) \sqrt{\cos^2(t) + 4 + \sin^2(t)} \, dt \\ &= \int_0^{\pi/2} \sin^2(t) \cos(t) \sqrt{5} \, dt = \frac{\sqrt{5}}{3} \sin^3(t) \Big|_0^{\pi/2} = \frac{\sqrt{5}}{3} (1 - 0) = \frac{\sqrt{5}}{3} \end{aligned}$$

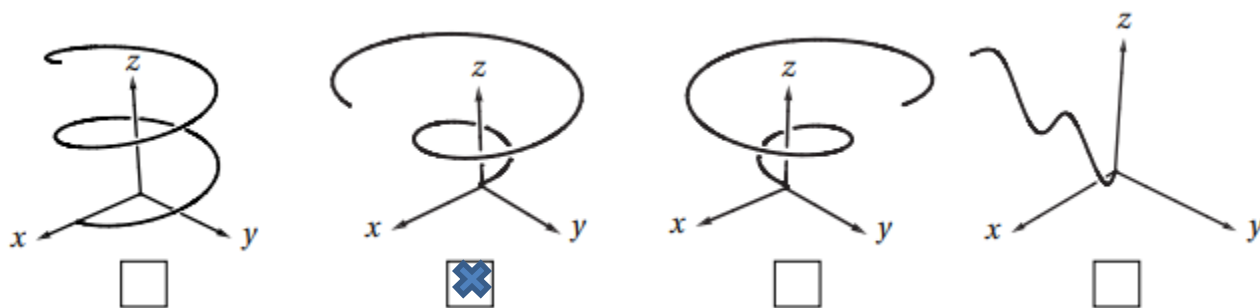
- b) Evaluate the integral $\int_C x^2 z \, dx$.

$$\begin{aligned} \int_C x^2 z \, dx &= \int_0^{\pi/2} \sin^2(t) \cos(t) \cos(t) \, dt = \int_0^{\pi/2} \sin^2(t) \cos^2(t) \, dt = \int_0^{\pi/2} \frac{1}{2} [1 - \cos(2t)] \frac{1}{2} [1 + \cos(2t)] \, dt \\ &= \frac{1}{4} \int_0^{\pi/2} [1 - \cos^2(2t)] \, dt = \frac{1}{4} \int_0^{\pi/2} \left[1 - \frac{1}{2} [1 + \cos(4t)] \right] \, dt = \frac{1}{4} \int_0^{\pi/2} \left[\frac{1}{2} - \frac{1}{2} \cos(4t) \right] \, dt \\ &= \frac{1}{4} \left(\frac{1}{2} t - \frac{1}{8} \sin(4t) \right) \Big|_0^{\pi/2} = \frac{1}{4} \left[\left(\frac{\pi}{4} - 0 \right) - (0 - 0) \right] = \frac{\pi}{16} \end{aligned}$$

- c) Evaluate the integral $\int_C x^2 z \, dy$.

$$\int_C x^2 z \, dy = \int_0^{\pi/2} \sin^2(t) \cos(t) 2 \, dt = \frac{2}{3} \sin^3(t) \Big|_0^{\pi/2} = \frac{2}{3} (1 - 0) = \frac{2}{3}$$

2. Mark the picture of the curve in $\mathbb{R}^3$ parameterized by $\mathbf{r}(t) = \langle t \cos(t), t \sin(t), t \rangle$ for $0 \leq t \leq 4\pi$.



3. Let C be the curve parameterized by $\mathbf{r}(t) = \langle \cos(t), \sin(t), 2t \rangle$ for $0 \leq t \leq 2\pi$.

a) Compute $\int_C z \, ds$.

We see that

$$\int_C z \, ds = \int_0^{2\pi} (2t) \sqrt{\sin^2(t) + \cos^2(t) + 4} \, dt = \int_0^{2\pi} (2t) \sqrt{5} \, dt = t^2 \sqrt{5} \Big|_0^{2\pi} = 4\pi^2 \sqrt{5}$$

b) Compute $\int_{-C} z \, ds$.

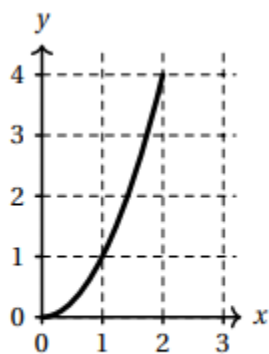
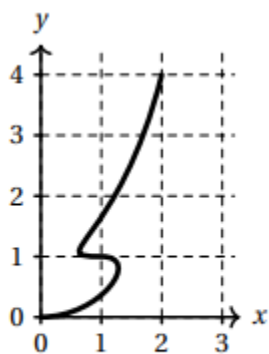
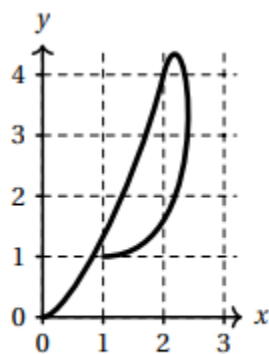
The direction of the curve has no influence on the value of a line integral with respect to arc length. We know that $\int_{-C} z \, ds = \int_C z \, ds = 4\pi^2 \sqrt{5}$

4. Find the mass of a thin wire in the shape of the curve $\mathbf{r}(t) = \langle \sin(t), 2\sin(t), \sqrt{5}\cos(t) \rangle$ where $0 \leq t \leq \pi$, if the wire has density function $\rho(x, y, z) = x$.

$$\begin{aligned} \text{mass} &= \int_C \rho(x, y, z) \, ds = \int_0^\pi \sin(t) \sqrt{\cos^2(t) + 4\cos^2(t) + 5\sin^2(t)} \, dt \\ &= \int_0^\pi \sin(t) \sqrt{5} \, dt = -\sqrt{5} \cos(t) \Big|_0^\pi = -\sqrt{5}(-1 - 1) = 2\sqrt{5} \end{aligned}$$

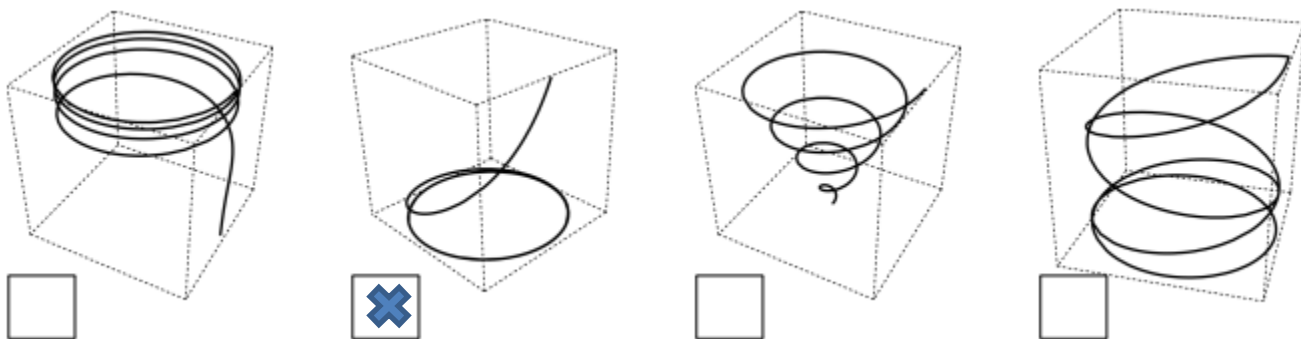
5. Suppose that $\mathbf{r}: [0, 2] \rightarrow \mathbb{R}^2$ is a parametric curve in the plane and that $\mathbf{r}(t)$ and $\mathbf{r}'(t)$ have the values given in the table at left. Check the box below the picture that could be a plot of $\mathbf{r}(t)$.

t	$\mathbf{r}(t)$	$\mathbf{r}'(t)$
0	(0, 0)	$\mathbf{i}$
1	(1, 1)	$-\mathbf{i}$
2	(2, 4)	$\mathbf{i} + 4\mathbf{j}$


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6. Let C be the oriented curve parameterized by $\mathbf{r}(t) = \langle \cos(t), \sin(t), e^t \rangle$ where $0 \leq t \leq 8\pi$.

a) Check the box next to the picture which best matches C .



b) Calculate the line integral $\int_C z \, dz$

$$\int_C z \, dz = \int_0^{8\pi} e^t e^t \, dt = \int_0^{8\pi} e^{2t} \, dt = \frac{1}{2} e^{2t} \Big|_0^{8\pi} = \frac{1}{2} (e^{16\pi} - 1)$$

c) Calculate the line integral $\int_{-C} z \, dz$.

We know that changing the orientation of the curve in a line integral with respect to x , y , or z will multiply the original result by -1 .

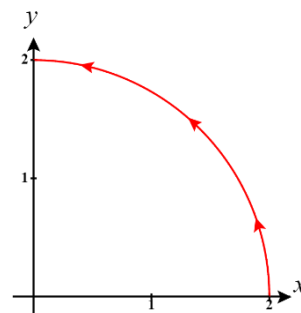
$$\text{Therefore, } \int_{-C} z \, dz = \int_C z \, dz = -\frac{1}{2} (e^{16\pi} - 1)$$

7. Suppose a wire can be modeled by the segment of the circle $x^2 + y^2 = 4$ from $(2, 0)$ to $(0, 2)$, when moving counterclockwise. If the linear density of this wire is measured by $f(x, y) = xy$ for any point on the wire, then set up an integral which represents the mass of this wire.

We know that the mass of the wire is given by $M = \int_C xy \, ds$ where C can be parameterized at

$\mathbf{r}(t) = \langle 2 \cos(t), 2 \sin(t) \rangle$ where $0 \leq t \leq \frac{\pi}{2}$. Therefore, we calculate that

$$\begin{aligned} M &= \int_C xy \, ds = \int_0^{\pi/2} 4 \cos(t) \sin(t) \sqrt{4 \cos^2(t) + 4 \sin^2(t)} \, dt \\ &= 8 \int_0^{\pi/2} \cos(t) \sin(t) \, dt = 4 \sin^2(t) \Big|_0^{\pi/2} \\ &= 4(1) - 4(0) \\ &= 4 \end{aligned}$$

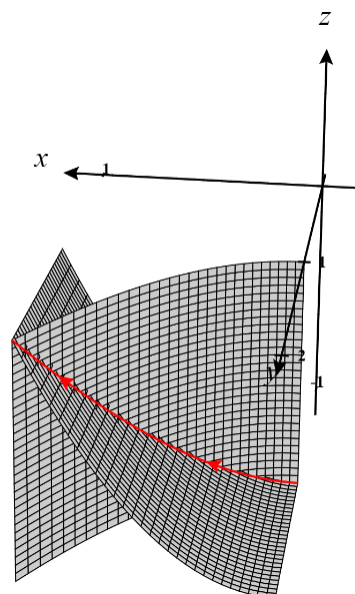


8. Let $f(x, y, z) = xy - xz$, and C be a curve which is the intersection of the surfaces $y = x^2 + 1$ and $z = x^2 - 1$ from $(0, 1, -1)$ to $(1, 2, 0)$. Compute the line integral $\int_C f(x, y, z) ds$.

We can see that the intersection curve C can be easily parameterized as $\mathbf{r}(t) = \langle t, t^2 + 1, t^2 - 1 \rangle$ where $0 \leq t \leq 1$.

Therefore, we compute that

$$\begin{aligned} \int_C f(x, y, z) ds &= \int_0^1 \left(t(t^2 + 1) - t(t^2 - 1) \right) \sqrt{1^2 + (2t)^2 + (2t)^2} dt \\ &= \int_0^1 2t \sqrt{8t^2 + 1} dt = \frac{1}{12} (1 + 8t^2)^{3/2} \Big|_0^1 \\ &= \frac{1}{12} (9)^{3/2} - \frac{1}{12} (1)^{3/2} = \frac{1}{12} (27 - 1) \\ &= \frac{13}{6} \end{aligned}$$



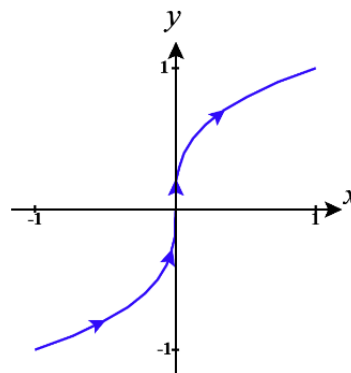
9. Evaluate the line integral $\int_C e^x dx$, where C is the arc of the curve $x = y^3$ from $(-1, -1)$ to $(1, 1)$.

We first parameterize C as $\mathbf{r}(t) = \langle t^3, t \rangle$ for $-1 \leq t \leq 1$.

Therefore, $\int_C e^x dx = \int_{-1}^1 e^{t^3} 3t^2 dt$

Let $u = t^3$ & $du = 3t^2 dt$. The bounds become $u(-1) = -1$ & $u(1) = 1$. So we get

$$\int_{-1}^1 e^{t^3} 3t^2 dt = \int_{-1}^1 e^u du = e^u \Big|_{-1}^1 = e^1 - e^{-1}$$



10. Consider the line integral $\int_C xye^{yz} dy$, where C is the arc of the curve parameterized by $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$ from $0 \leq t \leq 1$.

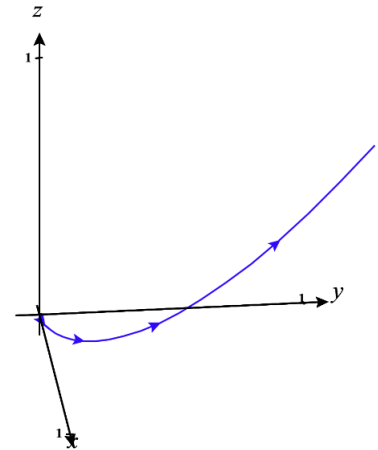
a) Evaluate $\int_C xye^{yz} dy$.

$$\int_C xye^{yz} dy = \int_0^1 (t)(t^2)e^{(t^2)(t^3)} 2t dt = \int_0^1 2t^4 e^{t^5} t dt$$

Let $u = t^5$ & $du = 5t^4 dt \rightarrow \frac{1}{5} du = t^4 dt$. The bounds become

$u(0) = 0$ & $u(1) = 1$. So we get

$$\int_0^1 2t^4 e^{t^5} t dt = \int_0^1 \frac{2}{5} e^u du = \frac{2}{5} e^u \Big|_0^1 = \frac{2}{5} e - \frac{2}{5}$$



b) Evaluate $\int_{-C} xye^{yz} dy$.

We know that changing the orientation of the curve in a line integral with respect to x , y , or z will multiply the original result by -1 .

$$\text{Therefore, } \int_{-C} xye^{yz} dy = \int_C xye^{yz} dy = -\left(\frac{2}{5}e - \frac{2}{5}\right)$$

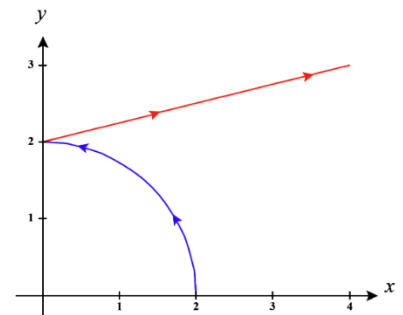
11. Evaluate the line integral $\int_C x^2 dx + y^2 dy$, where C consists of the arc of the circle $x^2 + y^2 = 4$ from $(2,0)$ to $(0,2)$ followed by the line segment from $(0,2)$ to $(4,3)$.

We will call the portion of C given by the circle C_1 with parameterization

$$\mathbf{r}_1(t) = \langle 2\cos(t), 2\sin(t) \rangle \text{ where } 0 \leq t \leq \frac{\pi}{2}.$$

We will call the portion of C given by the line segment C_2 with parameterization $\mathbf{r}_2(t) = \langle 4t, t+2 \rangle$ where $0 \leq t \leq 1$. We know that

$$\begin{aligned} \int_C x^2 dx + y^2 dy &= \int_{C_1} x^2 dx + y^2 dy + \int_{C_2} x^2 dx + y^2 dy = \int_{C_1} x^2 dx + \int_{C_1} y^2 dy + \int_{C_2} x^2 dx + \int_{C_2} y^2 dy \\ &= \int_0^{\pi/2} (4t)^2 4dt + \int_0^{\pi/2} (t+2)^2 dt + \int_0^1 (4t)^2 4dt + \int_0^1 (t+2)^2 dt \\ &= \frac{64}{3} t^3 \Big|_0^{\pi/2} + \frac{1}{3} (t+2)^3 \Big|_0^{\pi/2} + \frac{64}{3} t^3 \Big|_0^1 + \frac{1}{3} (t+2)^3 \Big|_0^1 \\ &= \frac{64\pi}{3 \cdot 8} + \left(\frac{1}{3} \left(\frac{\pi}{2} + 2 \right)^3 - \frac{1}{3} (2)^3 \right) + \frac{64}{3} + \left(\frac{1}{3} (3)^3 - \frac{1}{3} (2)^3 \right) \\ &= \frac{8\pi}{3} + \frac{1}{3} \left(\frac{\pi}{2} + 2 \right)^3 - \frac{8}{3} + \frac{64}{3} + 9 - \frac{8}{3} = \frac{8\pi}{3} + \frac{1}{3} \left(\frac{\pi}{2} + 2 \right)^3 + 25 \end{aligned}$$

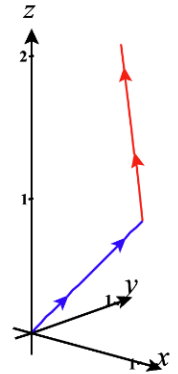


12. Evaluate the line integral $\int_C (y+z)dx + (x+z)dy + (x+y)dz$, where C consists of the line segments from $(0,0,0)$ to $(1,0,1)$ and from $(1,0,1)$ to $(0,1,2)$.

We will call the portion of C given by the first line segment C_1 with parameterization $\mathbf{r}_1(t) = \langle t, 0, t \rangle$ where $0 \leq t \leq 1$.

We will call the portion of C given by the line segment C_2 with parameterization $\mathbf{r}_2(t) = \langle -t+1, t, t+1 \rangle$ where $0 \leq t \leq 1$.

We know that



$$\int_C (y+z)dx + (x+z)dy + (x+y)dz = \int_{C_1} (y+z)dx + (x+z)dy + (x+y)dz + \int_{C_2} (y+z)dx + (x+z)dy + (x+y)dz$$

We calculate the line integral along each curve separately below.

$$\int_{C_1} (y+z)dx + (x+z)dy + (x+y)dz = \int_0^1 (0+t)dt + (t+t)0dt + (t+0)dt = \int_0^1 2tdt = t^2 \Big|_0^1 = 1$$

$$\begin{aligned} \int_{C_2} (y+z)dx + (x+z)dy + (x+y)dz &= \int_0^1 (t+t+1)(-1)dt + (-t+1+t+1)dt + (-t+1+t)dt \\ &= \int_0^1 (-2t-1)dt + (2)dt + (1)dt = \int_0^1 (-2t+2)dt = -t^2 + 2t \Big|_0^1 \\ &= -1+2 = 1 \end{aligned}$$

Therefore,

$$\begin{aligned} \int_C (y+z)dx + (x+z)dy + (x+y)dz &= \int_{C_1} (y+z)dx + (x+z)dy + (x+y)dz + \int_{C_2} (y+z)dx + (x+z)dy + (x+y)dz \\ &= 1+1 = 2 \end{aligned}$$